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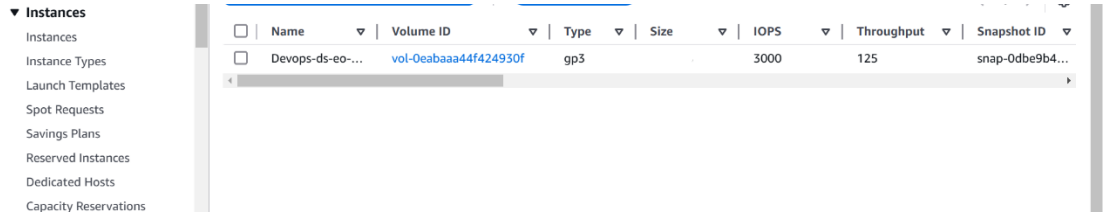
EBS Volume Extension of our EC2 Instance from 100 gb to 150 gb

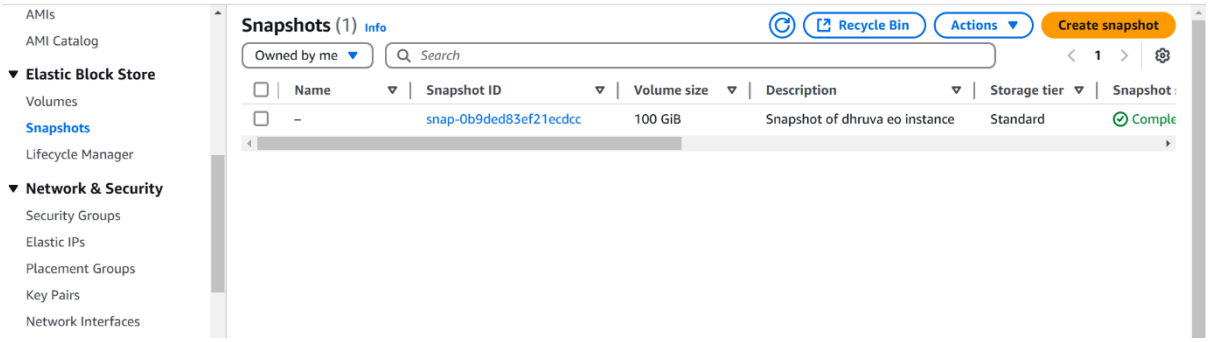
Background: The EC2 instance's EBS volume, originally sized at 100 GB, reached full capacity, leaving no space for new deployments. To address this issue, I suggested extending the EBS volume from 100 GB to 150 GB and shared the cost details. After receiving approval, I proceeded to extend the volume.

Objective: Increase the size of the EBS volume from 100 GB to 150 GB to accommodate future deployments and ensure uninterrupted operations.

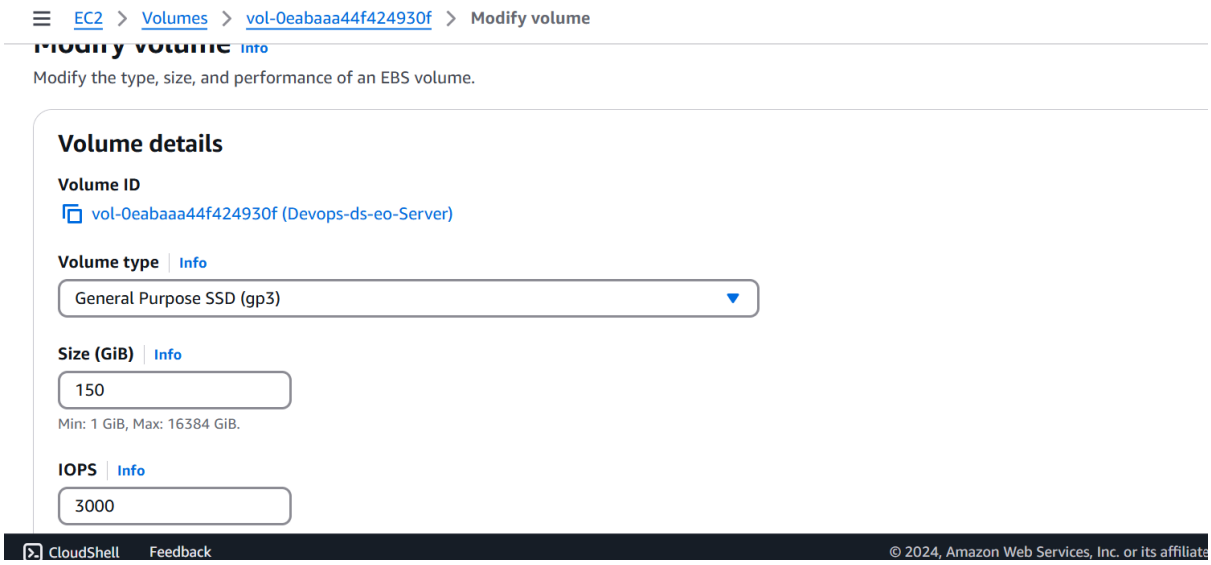


Steps Taken:

1. **Snapshot Creation:**
   - Created a snapshot of the current EBS volume as a precautionary measure.
   - Duration: Approximately 1 hour.



2. **Volume Modification:**
   - Accessed the AWS Management Console.
   - Modified the EBS volume size from 100 GB to 150 GB.
   - Duration: Approximately 1 hour to complete and reach the "in-use" state.



3. **Verified Volume Size:**
   - Logged into the EC2 instance.
   - Checked the volume size and confirmed that the original 100 GB was full.

```
[ec2-user@ip-10-98-80-87 krakend]$ df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             7.8G     0  7.8G   0% /dev/shm
tmpfs             3.1G   12M  3.1G   1% /run
/dev/nvme0n1p1    100G  100G  209M 100% /
tmpfs             7.8G  3.3M  7.8G   1% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs             1.6G     0  1.6G   0% /run/user/0
[ec2-user@ip-10-98-80-87 krakend]$
```

4. **Extended the Volume:**
   - Executed the following commands to extend the volume within the EC2 instance:
     1. `sudo lsblk` – To list block devices and identify the partition.
     2. `sudo growpart /dev/nvme0n1 1` – To extend the partition.
     3. `sudo lsblk` – To verify the partition extension.

```
sh-5.2$ sudo su
[root@ip-10-98-80-87 bin]# sudo lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0  150G  0 disk
├─nvme0n1p1   259:1    0  150G  0 part /
├─nvme0n1p127 259:2    0    1M  0 part
└─nvme0n1p128 259:3    0   10M  0 part /boot/efi
[root@ip-10-98-80-87 bin]#
```

4. `df -hT` – To check the filesystem and volume details.
5. `sudo xfs_growfs -d /` – To grow the filesystem and make use of the additional space.

5. **Validation:**
   - Verified that the EBS volume size was successfully extended to 150 GB.
   - Ensured the additional space was available for use within the EC2 instance.

```
[ec2-user@ip-10-98-80-87 ~]$ df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             7.8G     0  7.8G   0% /dev/shm
tmpfs             3.1G   12M  3.1G   1% /run
/dev/nvme0n1p1    150G  105G   46G  70% /
tmpfs             7.8G  3.3M  7.8G   1% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs             1.6G     0  1.6G   0% /run/user/0
[ec2-user@ip-10-98-80-87 ~]$
```

Outcome: The EBS volume was successfully extended from 100 GB to 150 GB, resolving the space constraint issue and enabling new deployments.